

Pivot Plan: From Breadboard to Lighting Circuits

The client does not offer a breadboard trainer. This is how the same system, the same flow and most of the same code move to a TESDA competency the client actually teaches.

Replaces The domain half of `build-plan.md`. Parts A and C stand as written
Status Proposal, to accept or reject before any code moves
Date 18 September 2026

1 The topic we should take

RECOMMENDED TOPIC

Install lighting fixtures and switch controls

TESDA EIM NC II · *Install Electrical Protective Devices for Distribution, Power, Lighting, Auxiliary, Lightning Protection and Grounding Systems* · **Learning Outcome 3**

One learning outcome, three exercises, one competency. Not the whole module.

#	EXERCISE	SOURCE IN THEIR MATERIAL	DEVICES
1	One lamp on a single-pole switch	Information Sheet 2.3-1	breaker, switch, lamp
2	Two lamps on a duplex switch	Job Sheet 2.3-1	breaker, 2-gang switch, 2 lamps
3	One lamp on two three-way switches	Information Sheet 2.3-1, three-way pages	breaker, 2× 3-way switch, lamp

Exercise 2 is their own culminating assessment for this outcome. Exercise 3 is the one students cannot debug without help, which is the entire reason our system exists.

2 Why this topic and not another

We looked at every practical sheet in the module. This one is the closest structural match to what we have already built, on five counts.

- **Same size.** Three to eight components per exercise, exactly like a breadboard circuit. Grounding and lightning protection are whole-building problems with no natural boundary; a lighting circuit has a beginning and an end.
- **Same mechanic.** The student joins terminals with conductors. We already know how to model a thing pushed into a connection point and turn it into a graph.
- **Same mathematics.** Netlist extraction and graph comparison carry over untouched, so Phases 11, 12 and 14 survive the pivot with no algorithmic change.
- **Faults are localizable.** Every fault lands on a named terminal in a named box, the way breadboard faults landed on a row.
- **It preserves our headline metric, and sharpens it.** See section 6.

By contrast, circuit breakers and panel boards (Learning Outcomes 1 and 2) are mostly identification and selection knowledge, which is a quiz rather than a diagnostic engine. Grounding and lightning protection have too few discrete, student-committed faults to build a 150-case corpus from.

3 What replaces the breadboard

Their material names it for us. Under **Conditions** for Learning Outcome 3 it lists *"Wiring booth / Simulated workplace"*. A TESDA wiring booth is a standard frame with fixed mounting positions. That fixed, known layout is what made the breadboard tractable, and it is what makes this tractable.

Assumed scope: wiring only

Boxes, conduit and fixture positions are placed by the exercise. The student pulls conductors and lands them on terminals. Every fault in section 5 is a wiring fault, so this targets all of the pedagogy without taking on box mounting and conduit bending. Flag this if the team disagrees. It is the one assumption that changes the schedule.

Addressing

BREADBOARD	LIGHTING BOOTH
<code>{ col: 1-63, row: 'A'-'J', half }</code>	<code>{ boxId, deviceId, terminalId }</code>
Hole	Terminal, and terminals are named , not spatial
Five-hole strip collapses to one node	Wire nut collapses its conductors to one node
Component spans two holes	Conductor spans two terminals

Named terminals are strictly easier than a coordinate grid. There is no raycasting, no nearest-hole maths and no span validation, so Phase 10 gets simpler.

Device model

Each device declares its terminals and its internal connectivity, which may depend on state. This small table is the whole domain; adding a device later is one row.

DEVICE	TERMINALS	INTERNAL CONNECTION
Breaker	<code>bus</code> , <code>load</code>	closed when on
Single-pole switch	<code>line</code> , <code>load</code>	closed when up
Three-way switch	<code>common</code> , <code>traveler1</code> , <code>traveler2</code>	<code>common</code> to exactly one traveler
Duplex switch	two independent single-pole sets	per gang
Receptacle	<code>line</code> , <code>neutral</code> , <code>ground</code>	always, to its blades
GFCI	<code>line-in</code> , <code>load-out</code> , neutrals, <code>ground</code>	trips on imbalance
Lamp	<code>t1</code> , <code>t2</code>	through the filament, a load

4 The verdict becomes three checks, not one

The breadboard had one verdict: does the netlist match. A lighting circuit gives us three, and they cross-check each other.

Check 1 — behaviour, by truth table

Evaluate connectivity in every switch state and compare against the reference behaviour. For exercise 3 that is four states.

SWITCH 1	SWITCH 2	LAMP
up	up	on
up	down	off
down	up	off
down	down	on

Check 2 — safety

Independent of whether the lamp lights: is the switch in the ungrounded conductor, is the equipment ground continuous end to end, is there any hot-to-neutral or hot-to-ground short, is the breaker rating within the conductor's ampacity.

Check 3 — match

Does the student's netlist match the instructor's captured reference, up to isomorphism.

Why three beats one

A circuit can pass Check 1 and fail Check 3, which means the student found a different but working wiring. That combination is our false-positive guard: when it fires, the comparator is wrong, not the student. The breadboard had no such independent signal.

It also shrinks Phase 16. We do not need modified nodal analysis. We need graph reachability per switch state plus short detection, which is a morning's work rather than four days.

5 Fault taxonomy

Tier 1 — intrinsic, no answer key, runs in the browser

FAULT	WHY IT MATTERS	AUTHORITY
Switched neutral	Fixture stays live when off. Kills people changing bulbs	NEC 404.2(B)
Hot and neutral reversed	Shell of the lampholder is live	NEC 200.11
Equipment ground missing or broken	No fault path, so the breaker never trips	NEC 250.4
Hot-to-neutral short	Dead short across the branch	—
Hot-to-ground short	Same, through the ground path	—
Unterminated conductor	Live copper loose in a box	—
Double-tapped breaker	Two conductors on a one-conductor lug	NEC 110.14
Breaker rating over conductor ampacity	Conductor cooks before the breaker opens	NEC 240.4
Wire below 2.0 mm ² TW on lighting	Their own stated minimum	Info Sheet 2.3-1 §G
Switch not at 1.5 m	Their own stated mounting height	Info Sheet 2.3-1 §F

Tier 2 — needs the reference, runs on the API

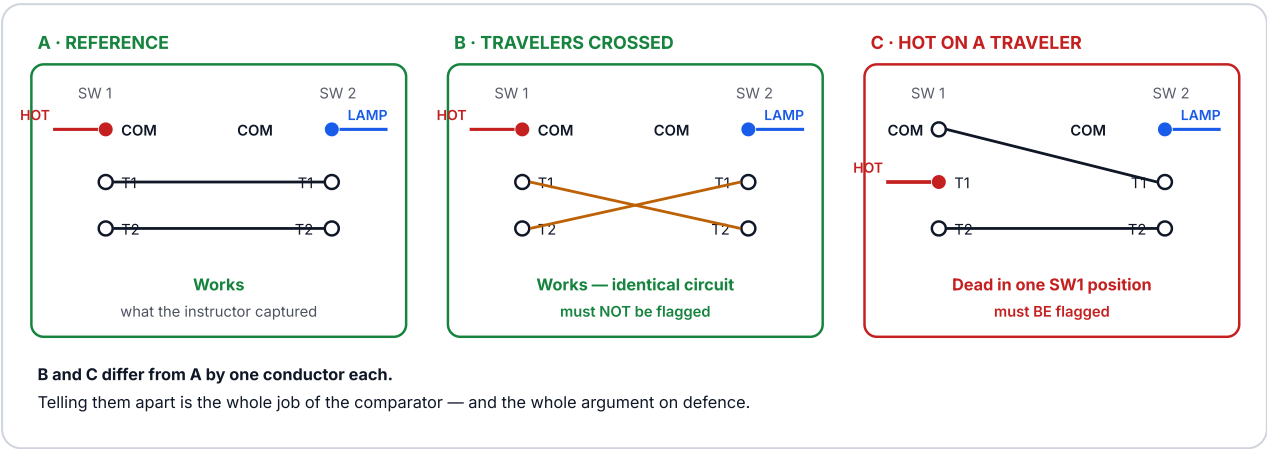
FAULT	NOTES
Hot feed on a traveler instead of the common	Lamp works in one position of the first switch, dead in the other. The classic three-way failure
Switch leg on a traveler instead of the common	Same symptom, other end of the run
Both travelers on one terminal	Second traveler does nothing; lamp dead in one position
Lamp on the wrong gang of the duplex switch	Both lamps work, wrong switch controls each
Device on the wrong branch circuit	Load schedule violation

Ten Tier 1 rules and five Tier 2 classes. The existing plan budgeted for eight Tier 1 rules, so Phase 13 is the same size. Every one of these cites a code article or their own material, which is a stronger footing than the breadboard taxonomy ever had.

6 The headline metric survives, and sharpens

The existing plan's most important number is that a correct circuit built differently must produce zero faults. On a breadboard that meant the same circuit built on different rows. Here it means something better, because three-way circuits contain a real isomorphism that students hit constantly.

The two travelers are interchangeable. It does not matter which traveler terminal on the first switch connects to which on the second. Both wirings are electrically identical, both work, and students produce both.



The test case that matters. A comparator matching conductors by terminal name flags B as a fault. It must not. It must still catch C, which looks almost identical on the board.

Other genuine isomorphisms to seed the negative corpus with: which of several neutral conductors lands on which neutral terminal, splice ordering inside a box, and which physical conduit path a conductor takes between the same two terminals.

7 What carries over untouched

PART	PHASES	CARRIES OVER
A — Student shell	0–8	Entirely. Auth, classes, dashboard, settings and the instructor shell do not know what the domain is. Only Phase 7's 3D content changes, not its layout
B — Engine	11, 12, 14, 15, 17–20	Netlist extraction, Learn Mode, the comparator, fault presentation, telemetry, analytics, PWA and the evaluation harness. No algorithmic change
C — Tasks, grading, AI	22–25	Entirely. Deadlines, timers, auto-grading, the release gate and the AI Coach are domain-agnostic

Roughly three quarters of the existing plan is unaffected. The pivot touches four phases.

8 What changes

PHASE	WAS	BECOMES	EFFORT
9	Board geometry, 830 holes, union-find over strips	Booth topology, named terminals, union-find over splices	4d → 3d
10	Drag components, raycast to holes, span validation	Pull conductors, land on named terminals, per-device rules	6d → 5d
13	Eight intrinsic breadboard rules	Ten code-based rules from section 5	4d → 4d
16	Modified nodal analysis, Gaussian elimination	Switch-state reachability and short detection	4d → 2d
new	—	Truth-table verdict and the three-check reconciliation	— → 2d

Net: Part B drops from 50 to 46 days. The saving is real but small; do not spend it in advance.

One addition worth making

Self-Check 2.3-1 is five questions with an answer key, already written. Wiring it in as a short pre-quiz costs about a day and gives the evaluation chapter a knowledge score to sit beside the performance score. Optional, and easy to cut.

9 Timeline

PART	PHASES	DAYS
A — Student shell	0–8	35
B — Engine	9–20	46
C — Tasks, grading, AI	22–25	19
Evaluation	21	10
Total		~110 days

Within two days of the breadboard plan, so the pivot is not a schedule event. **Phases 0 to 4 are already built and are unaffected.** No work is thrown away.

10 Risks the pivot introduces

RISK	MITIGATION
3D booth is more complex than a flat board: a panel, boxes at height, devices with several terminals	Wiring-only scope; boxes pre-placed; camera snaps to one box at a time rather than free orbit
Conductor routing looks like spaghetti in 3D	Route conductors along conduit automatically. The student picks terminals, not paths. The path was never the assessment
Team has less domain intuition for wiring faults than for breadboards	Every fault in section 5 cites a code article. Have an electrician review the corpus once, before Phase 20
Panel objects that the truth table alone is enough, making the comparator redundant	It is not. The truth table cannot tell a switched neutral from a correct circuit, because the lamp behaves identically. Lead with that example

11 What we need to decide

1. **Wiring-only scope** — accept, or add box mounting and conduit running.
2. **Three exercises or two** — dropping exercise 1 loses the gentle on-ramp but saves little, since it shares all its machinery with exercise 2.
3. **Self-Check quiz in or out.**

Nothing else about the project changes.